

Time Series Modelling Case Study

1. Introduction

In this case study, we will conduct a time series analysis on the oil prices dataset. The dataset contains daily oil prices observations from September 2024 to February 2026. The primary objectives of this analysis are to preprocess the data, perform stationary tests, model the oil price time series, evaluate model performance, and generate future forecasts with confidence intervals.

Two modelling approaches are employed: the classical Autoregressive Moving Average (ARMA) model, which serves as the baseline statistical approach, and the Long Short-Term Memory (LSTM) neural network, which represents a deep learning alternative capable of capturing complex non-linear temporal dependencies. Both models are evaluated using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), and their forecasts are compared and discussed.

2. Exploratory Data Analysis

The dataset consists of 500 daily oil price observations with no missing values and duplicates, which means that the data is clean and ready to use in further steps. The following table summarises the key statistics from the dataset.

Statistic	Value
Observations	500
Start Date	21 September 2024
End Date	2 February 2026
Minimum Price	~\$16
Maximum Price	~\$160
Mean Price	~\$85
Missing Values	0

Table 1: Summary Statistics

2.1 Daily Oil prices

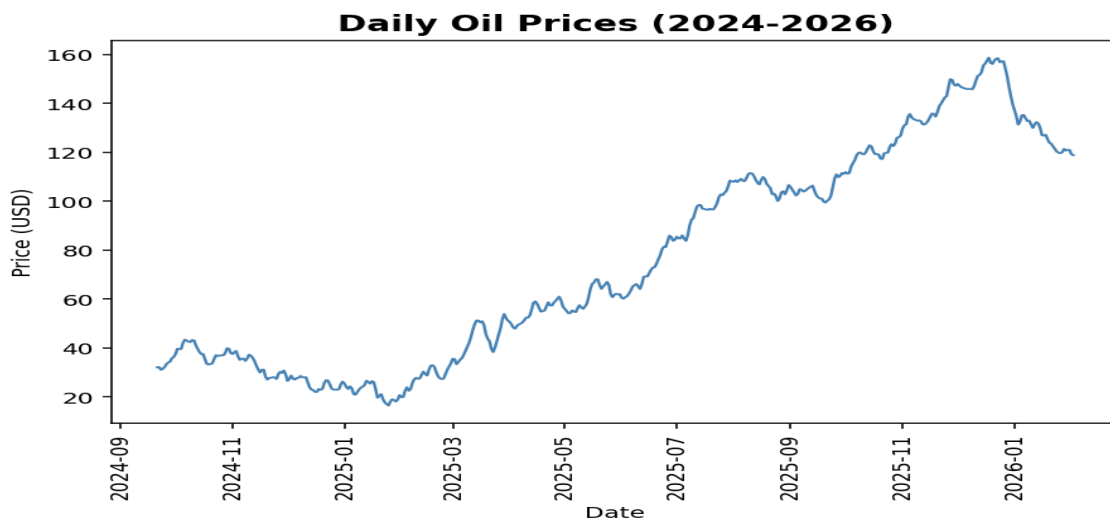


Figure 1: Daily Oil Prices

Figure 1 shows the daily oil prices from September 2024 to February 2026. The oil prices show a strong upward trend followed by a sharp decline in the final weeks of the observation period.

2.2 Rolling Mean and Standard Deviation

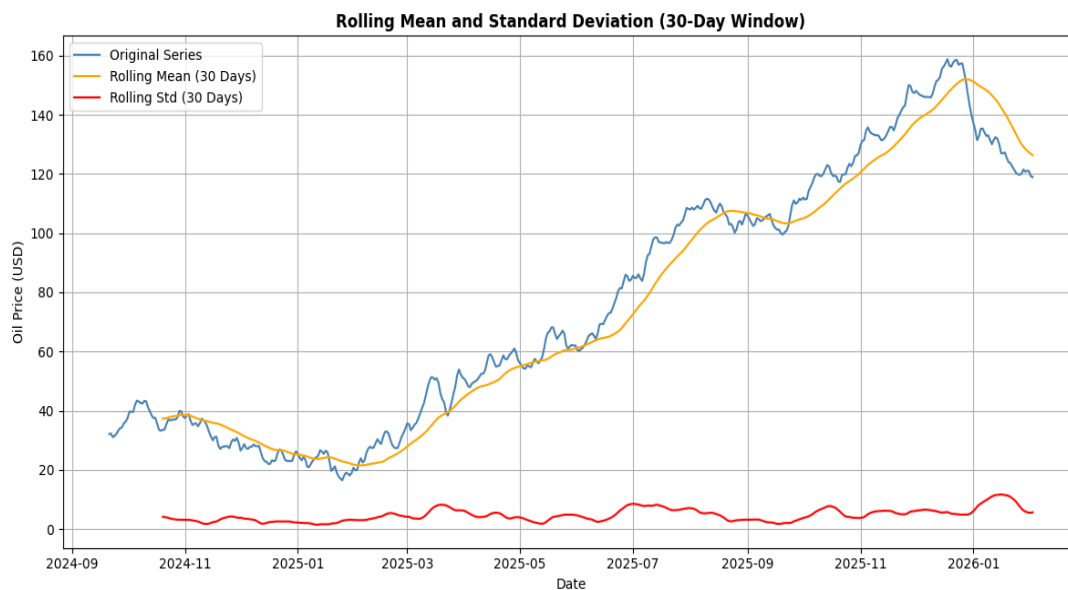


Figure 2: Rolling Mean and Standard Deviation

Figure 2 shows the rolling mean and standard deviation of oil prices computed over a 30-day window. Just like oil prices rolling mean follows the same trajectory, i.e. rising steadily all the way till the start of 2026 and then dropping sharply after that. Rolling standard deviation remains constant throughout the series, which means that oil prices were not getting irregular or inconsistent over time. The non-constant rolling mean provides a visual representation that the series is not stationary, and we need to apply techniques to make it stationary.

2.3 Price Distribution

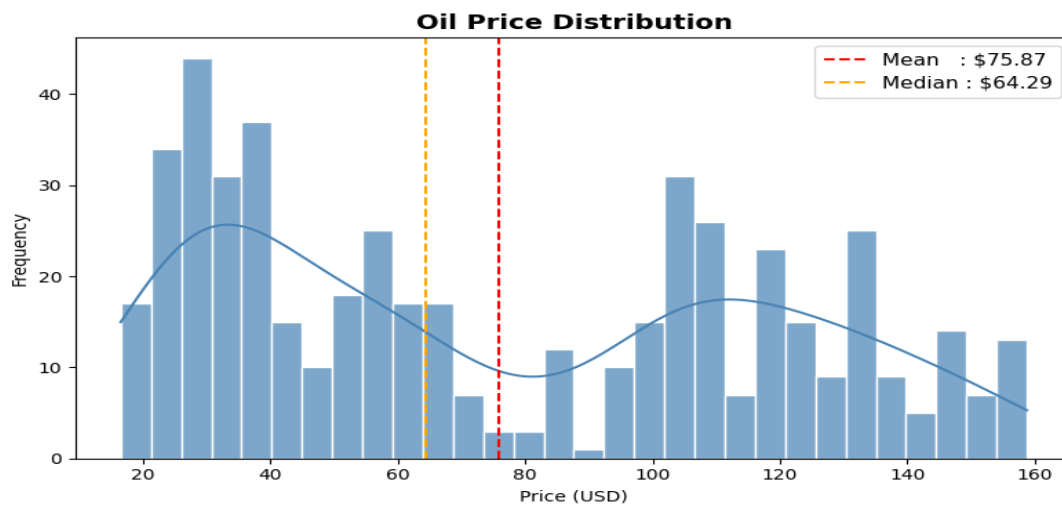


Figure 3: Oil Price Distribution

The histogram shows that the oil prices did not stay around one level throughout the dataset, but instead they clustered around two very different price ranges: a lower range of around \$25–\$35 and a higher range of around \$100–\$110. This reflects the journey the prices took over the observation period, starting low and rising significantly over time. The gap between mean and median tells us the data is skewed, and a single average does not tell the full story, which is exactly why models like ARMA and LSTM are needed to properly capture how prices evolved over time.

2.4 Seasonal Decomposition

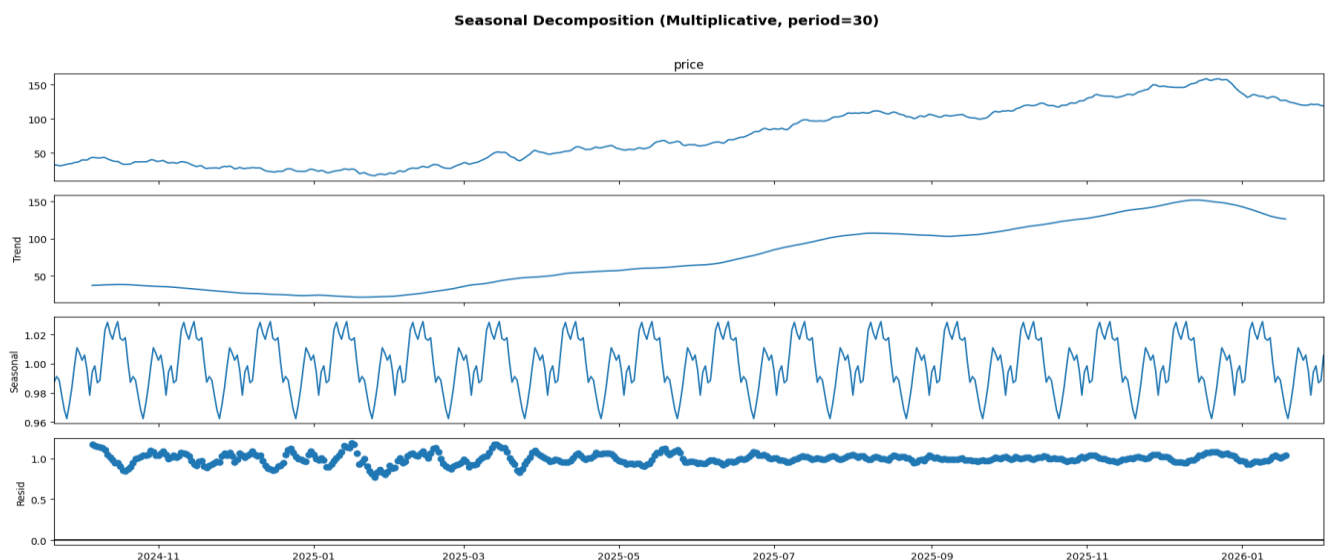


Figure 4: Seasonal Decomposition

The decomposition showed a weak seasonal component with an amplitude ranging between 0.96 and 1.02, which suggests that seasonality has a minimal influence on the overall price. The dominant component is the trend, which accounts for the majority of the variation in the series. Since trend is the major component, it means that ARIMA is enough for the analysis, rather than SARIMA, which is useful for seasonal data.

3. Stationary Tests

The initial AutoCorrelation Function (ACF) and Partial AutoCorrelation Function (PACF) plots justify the need to make the data stationary, fitting any model.

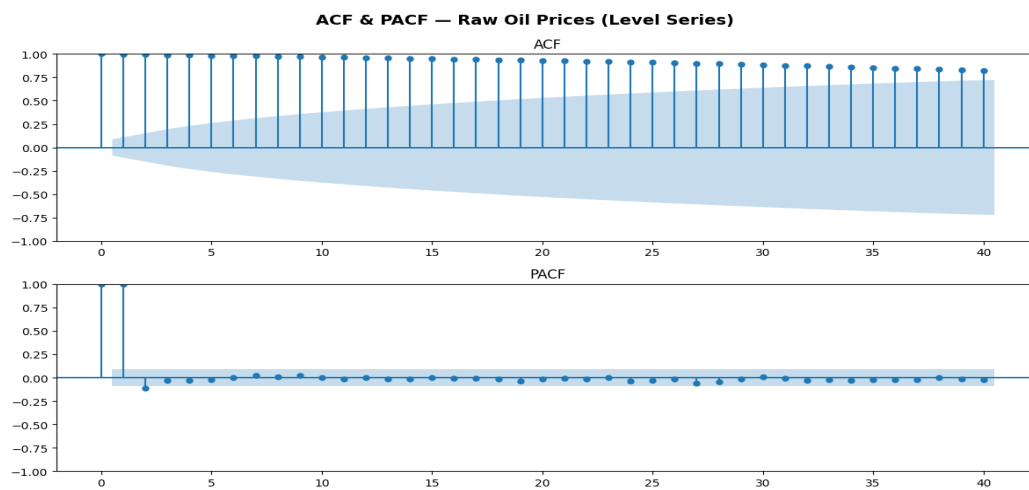


Figure 5: ACF and PACF plots

Figure 5 shows that ACF decays extremely slowly, which provides strong visual evidence for non stationary data. The slow decay also proves that the series has a long memory driven entirely by its upward trend rather than any cyclic pattern. The PACF also support this finding, showing only a dominant spike before cutting off sharply. Then, the Augmented Dickey-Fuller Test (ADF) was performed, which further justifies that the series has a unit root and is therefore non-stationary.

To address this issue, data was first log-transformed in order to stabilise the growing variance and then to remove the upward trend, first-order differencing was applied. After that ADF test confirmed the data is stationary.

4. ARIMA Modelling

Grid search was performed to select the best model parameters. Each model was evaluated using the Akaike Information Criterion (AIC), which penalises model complexity and guards against overfitting. The model with the lowest AIC was selected as the best model.

Parameter	Range Tested	Best Value
p (AR order)	0–8	2
d (differencing)	0–2	2
q (MA order)	0–8	7
AIC (best model)	—	1429.36
Total models tested	—	243

Table 2: ARIMA Parameters

The best model we got was ARIMA(2,2,7) with an AIC of 1429.36. The value of $d = 2$ indicates that the series required second-order differencing to achieve stationarity. The AR(p) order of 2 suggests that the current price depends on the two preceding values, while the MA(q) order of 7 indicates that forecast errors from the previous seven periods influence the current value.

The model achieved an RMSE of 1.89, indicating that, on average, the model's predictions deviated from the actual oil prices by approximately \$1.89. This is a strong result suggesting the model captured the underlying trend and short-term dynamics effectively.

4.1 Residual Diagnostics

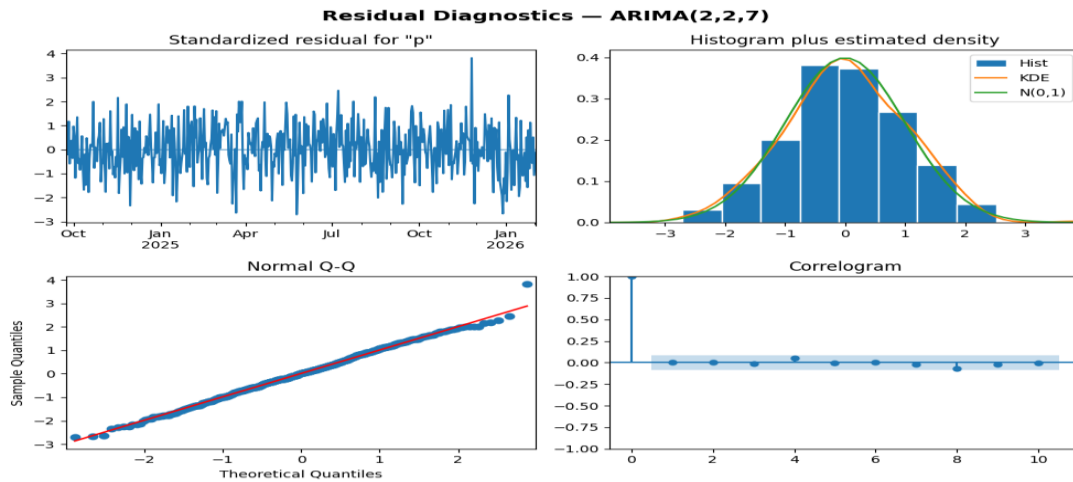


Figure 6: Residual Diagnostics

Figure 6 shows that the model has done a good job of capturing the underlying structure of the oil price series. The standardised residuals plot (top left) shows residuals fluctuating randomly around zero with no obvious pattern or trend, which is exactly what we want; it suggests the model has not missed any systematic information in the data. The histogram (top right) shows the residuals closely follow a normal distribution, confirmed by the KDE and $N(0,1)$ curves aligning well, indicating the model errors are approximately normally distributed. The Q-Q plot (bottom left) shows points lying closely along the diagonal red line, further confirming normality of residuals, with only minor deviations at the extremes. Finally, the correlogram (bottom right) shows no significant autocorrelation in the residuals beyond lag 0, meaning the model has successfully removed the temporal dependencies from the data. Overall, these diagnostics confirm that ARIMA(2,2,7) is a well-fitted model whose residuals behave like white noise.

4.2 24 Month Forecast

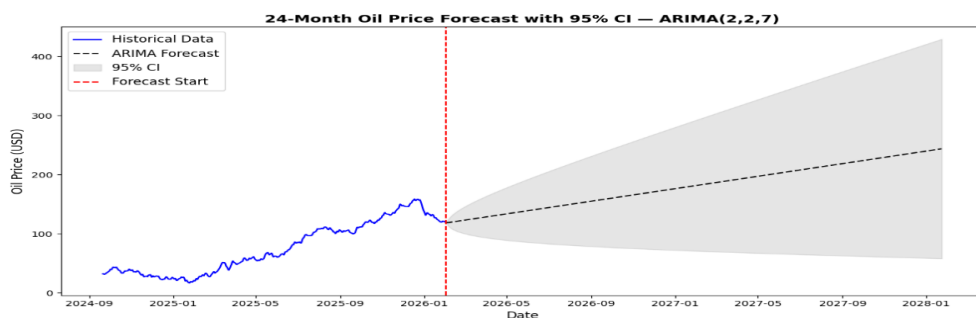


Figure 7: 24 Month Forecast of ARIMA

The forecast connects smoothly from the historical data and rises at a steady pace, because of $d=2$, the model has learned the accelerating upward trend in the training data and extrapolates it forward. The 95% confidence intervals widen substantially over time, which is statistically expected. The further into the future we forecast, the less certain we can be. By 2028, the confidence interval spans from approximately \$60 to \$430, reflecting the high degree of uncertainty over a 24-month horizon for a volatile commodity like oil. While the upward forecast is consistent with the dominant trend observed in the training data, it is important to note that ARIMA cannot account for sudden market shocks, geopolitical events, or structural breaks that could dramatically alter the price trajectory.

5. LSTM Modelling

5.1 Model Architecture

The Long Short-Term Memory (LSTM) neural network is a type of recurrent neural network (RNN) specifically designed to learn long-range temporal dependencies through its gating mechanism (Hochreiter & Schmidhuber, 1997). Unlike traditional feedforward networks, LSTM maintains a cell state that allows information to persist across many time steps, making it well-suited for financial time series modelling.

The architecture employed in this analysis consists of two stacked LSTM layers, each with a configurable number of units, followed by a Dense output layer. Dropout regularisation with a rate of 0.2 was applied after each LSTM layer to prevent overfitting. This stacked architecture allows the first layer to learn short-term patterns and the second layer to learn longer-term dependencies — a design motivated by the hierarchical feature learning capabilities of deep networks.

5.2 Model Training

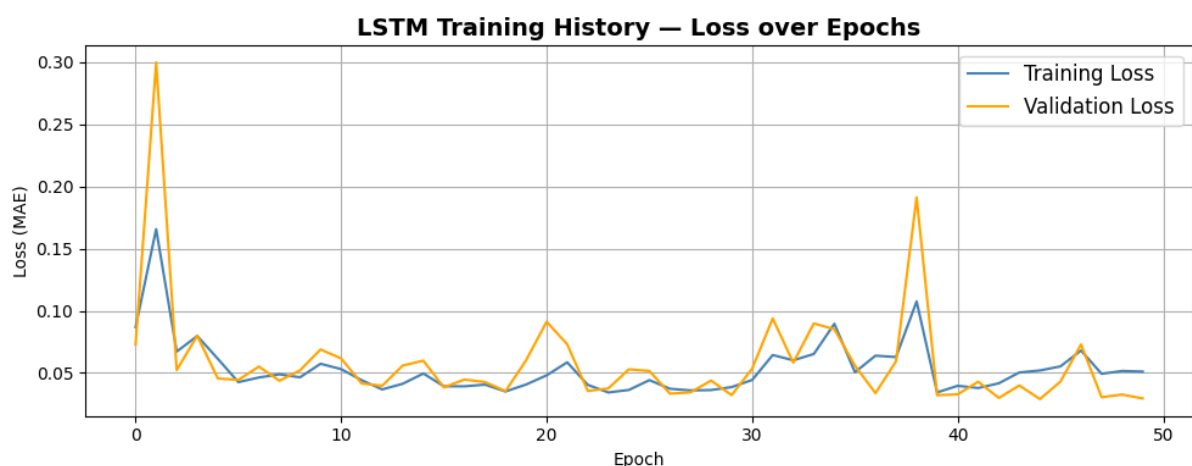


Figure 8: LSTM Training History

The training loss from Figure 8 shows that the LSTM loss decreased over 50 epochs. The rapid initial decline indicates that the model learns the core patterns in the data quickly. After the first few epochs, both losses remain relatively flat and track each other closely, which means that the model is not overfitting. The occasional spikes in validation loss are normal as the model is adjusting its weight in response to difficult batches. Overall, the training history suggests a well-behaved and stable model.

5.3 Hyperparameter Tuning

Hyperparameter tuning was conducted using a grid search over three key parameters: LSTM units, batch size, and learning rate, yielding 27 candidate configurations. Each configuration was trained using the Adam optimiser with Mean Absolute Error (MAE) as the loss function and evaluated on the validation set.

Parameter	Values Tested	Best Value
LSTM units	32, 50, 100	50
Batch size	16, 32, 64	16
Learning rate	0.01, 0.001, 0.0001	0.01
Total combinations	27	—
Selection criterion	—	0.014 (lowest validation loss)

Table 3: Hyperparameters of LSTM

Following hyperparameter tuning, the LSTM model achieved an RMSE of 4.13 and MAE of 3.06, representing an average prediction error of approximately 2–3% relative to the actual test prices, indicating strong predictive performance.

5.4 24 Month Forecast

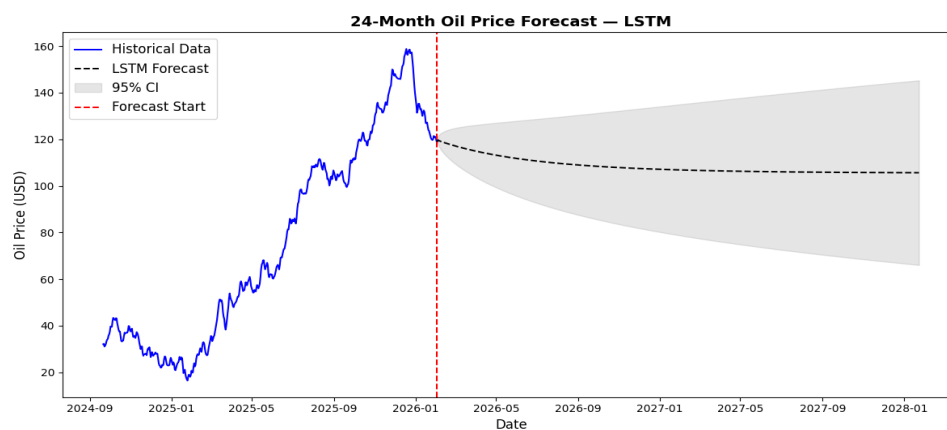


Figure 9: 24 Month Forecast of LSTM

The 24-month LSTM forecast was generated using recursive multi-step forecasting, where each predicted value is fed back as input for the next prediction step. The forecast shows prices declining slightly from \$119 to approximately \$105 before stabilising over the forecast horizon.

The stabilisation behaviour of the LSTM forecast is a known characteristic of recursive multi-step forecasting (Taieb et al., 2012). As the forecast horizon extends, the input window is increasingly populated by predicted rather than real values. After approximately 60 steps, equivalent to the lookback window, the model receives no real observations, causing it to

converge to a stable equilibrium value. This is not a modelling error but a limitation of the recursive forecasting strategy.

6. Model Comparison

ARIMA projects a continued upward trend reaching approximately \$240 by 2028, while LSTM, on the other hand, predicts a mild decline from \$119 to approximately \$105 before stabilising, reflecting the sharp price drop it observed in its 60-day input window just before the forecast start. LSTM operates as a black box, making it difficult to explain individual predictions, while ARIMA offers great interpretability. ARIMA outperformed LSTM with an RMSE of 1.86 compared to the LSTM's RMSE of 4.13 and MAE of 3.06. This suggests that for this dataset, the linear structure captured by ARIMA was sufficient to model the short-term price dynamics more accurately than the LSTM. This is not entirely surprising as LSTM typically requires large amounts of data to outperform classical statistical models, and a 500-day dataset may not provide enough observations for the neural network to fully learn the underlying patterns.

7. Future Improvements

- Instead of univariate modelling, multivariate modelling should be explored by adding more external features, such as USD exchange rates, geopolitical risks etc., since LSTM is more suitable for multivariate inputs.
- The dataset used in the modelling is very small (only 500 observations). For better results from machine learning and deep learning models, a large dataset should be considered in future.
- Hybrid models combining ARIMA and LSTM could be investigated. A common approach (Zhang, 2003) uses ARIMA to model the linear components of the series and LSTM to capture the residual non-linear patterns, potentially outperforming both models individually.

References

Hochreiter, S. & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735–1780.

Taieb, S.B., Bontempi, G., Atiya, A.F. & Sorjamaa, A. (2012). A review and comparison of strategies for multi-step ahead time series forecasting based on the NN5 forecasting competition. *Expert Systems with Applications*, 39(8), 7067–7083.

Zhang, G.P. (2003). Time series forecasting using a hybrid ARIMA and neural network model. *Neurocomputing*, 50, 159–175.